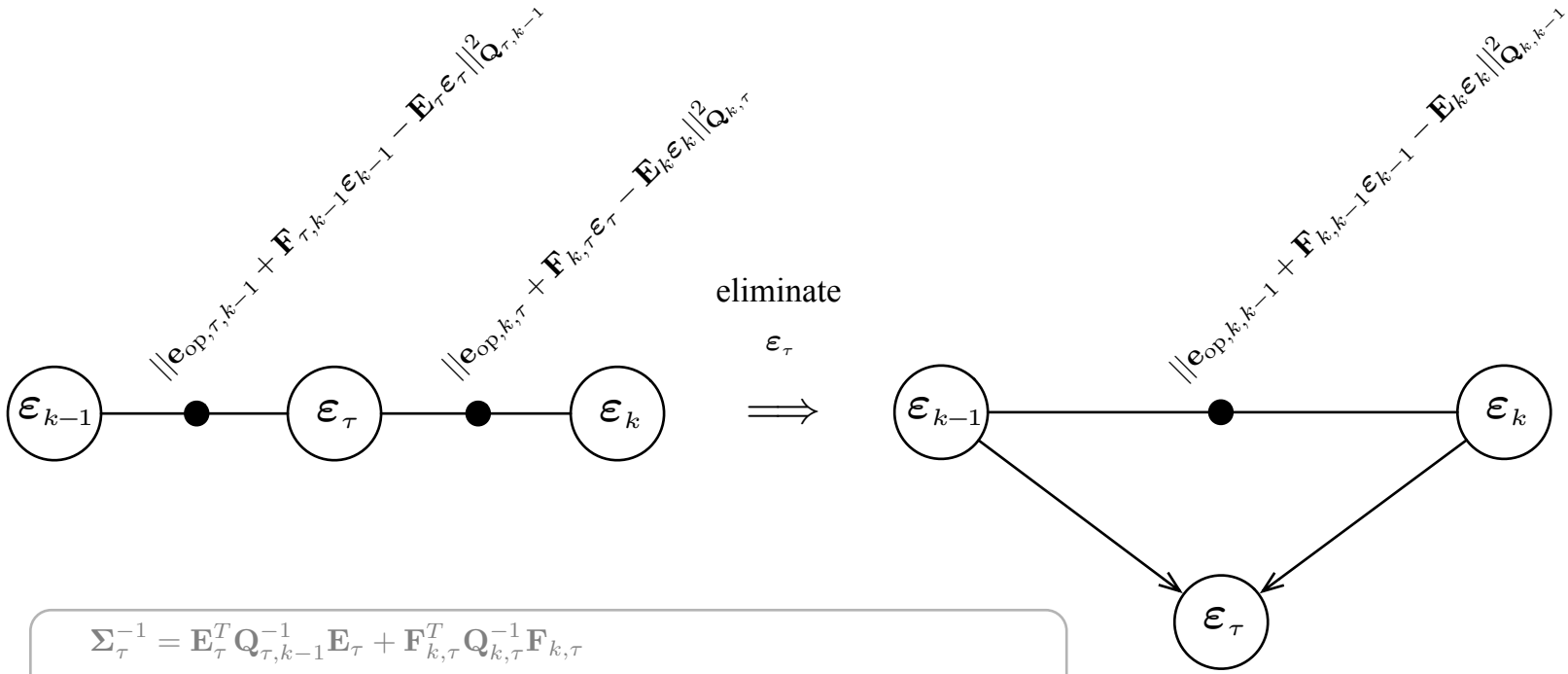

$$\boldsymbol{\Sigma}_\tau^{-1} = \mathbf{E}_\tau^T\mathbf{Q}_{\tau,k-1}^{-1}\mathbf{E}_\tau + \mathbf{F}_{k,\tau}^T\mathbf{Q}_{k,\tau}^{-1}\mathbf{F}_{k,\tau}$$

$$\boldsymbol{\Sigma}_\tau^{-1}\boldsymbol{\mu}_\tau = \mathbf{E}_\tau^T\mathbf{Q}_{\tau,k-1}^{-1}\left(\mathbf{F}_{\tau,k-1}\boldsymbol{\varepsilon}_{k-1} + \mathbf{e}_{\mathrm{op},\tau,k-1}\right) + \mathbf{F}_{k,\tau}^T\mathbf{Q}_{k,\tau}^{-1}\left(\mathbf{E}_k\boldsymbol{\varepsilon}_k - \mathbf{e}_{\mathrm{op},k,\tau}\right)$$